SCREENSHOTS

1)MULTI-CONTAINER SUPERVISION

```
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ mkdir -p rootfs-s1 rootfs-s2
cp -a rootfs/bin rootfs/lib rootfs/lib64 rootfs/memory_hog rootfs-s1/
cp -a rootfs/bin rootfs/lib rootfs/lib64 rootfs/memory_hog rootfs-s2/
mkdir -p rootfs-s1/proc rootfs-s2/proc
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine start s1 rootfs-s1 /memory_hog --soft-mib 2000
 --hard-mib 3000
sudo ./engine start s2 rootfs-s2 /memory_hog --soft-mib 2000 --hard-mib 3000
sudo ./engine ps
[sudo] password for abhijeet:
OK started id=s1 pid=17102 soft=2000MiB hard=3000MiB log=logs/s1.log
^C
OK started id=s2 pid=17315 soft=2000MiB hard=3000MiB log=logs/s2.log
^C
ID      PID     STATE   SOFT    HARD    START   ROOTFS  COMMAND REASON  LOG
s1      17102   running 2000MiB 3000MiB 2026-04-20_17:01:37     rootfs-s1       /memory_hog     none    logs/s1.log
s2      17315   running 2000MiB 3000MiB 2026-04-20_17:05:27     rootfs-s2       /memory_hog     none    logs/s2.log
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$
```

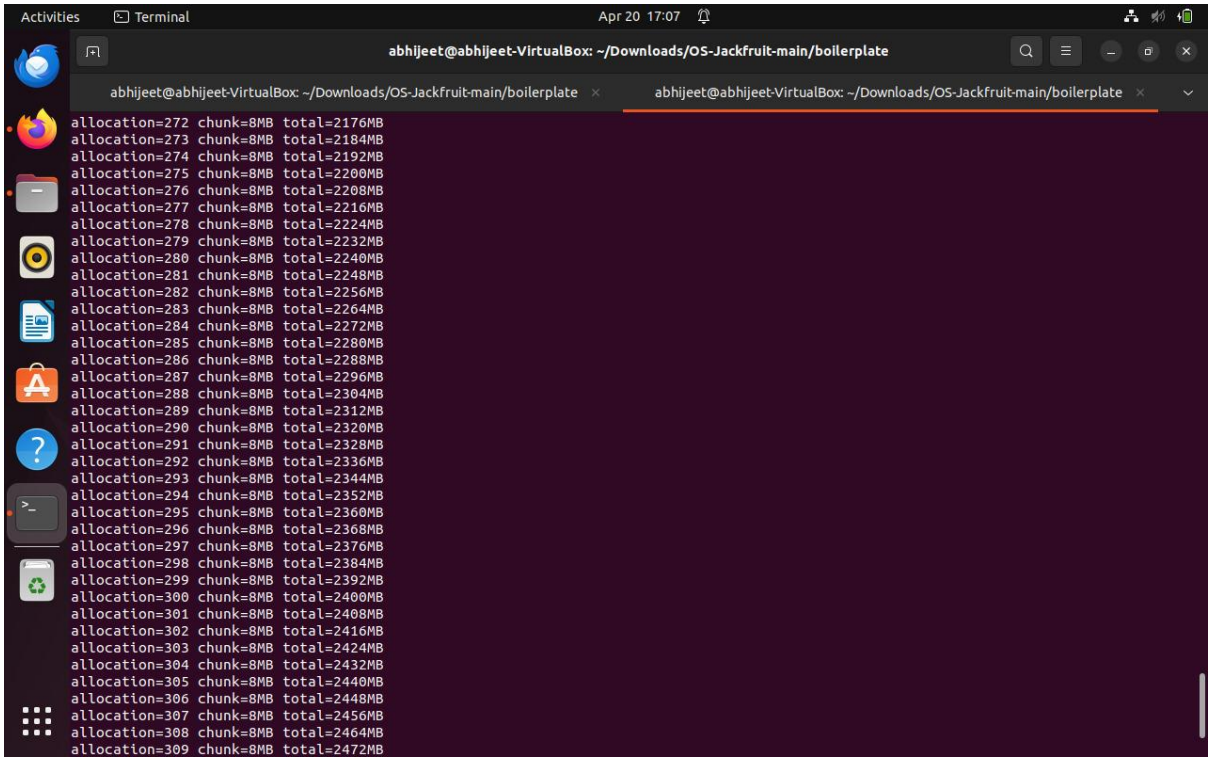
2)META DATA TRACKING

```
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine ps
ID      PID     STATE   SOFT    HARD    START   ROOTFS  COMMAND REASON  LOG
s1      17102   running 2000MiB 3000MiB 2026-04-20_17:01:37     rootfs-s1       /memory_hog     none    logs/s1.log
s2      17315   running 2000MiB 3000MiB 2026-04-20_17:05:27     rootfs-s2       /memory_hog     none    logs/s2.log
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$
```

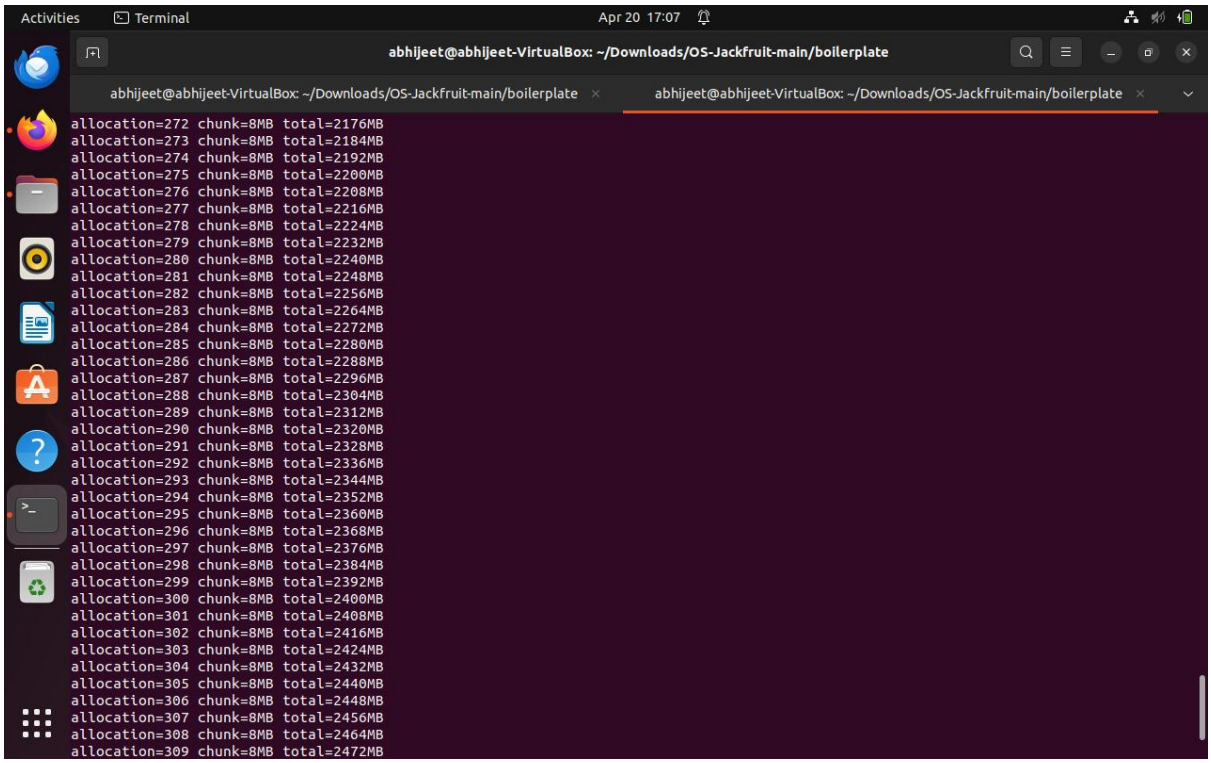
3.BOUNDED BUFFER LOGGING

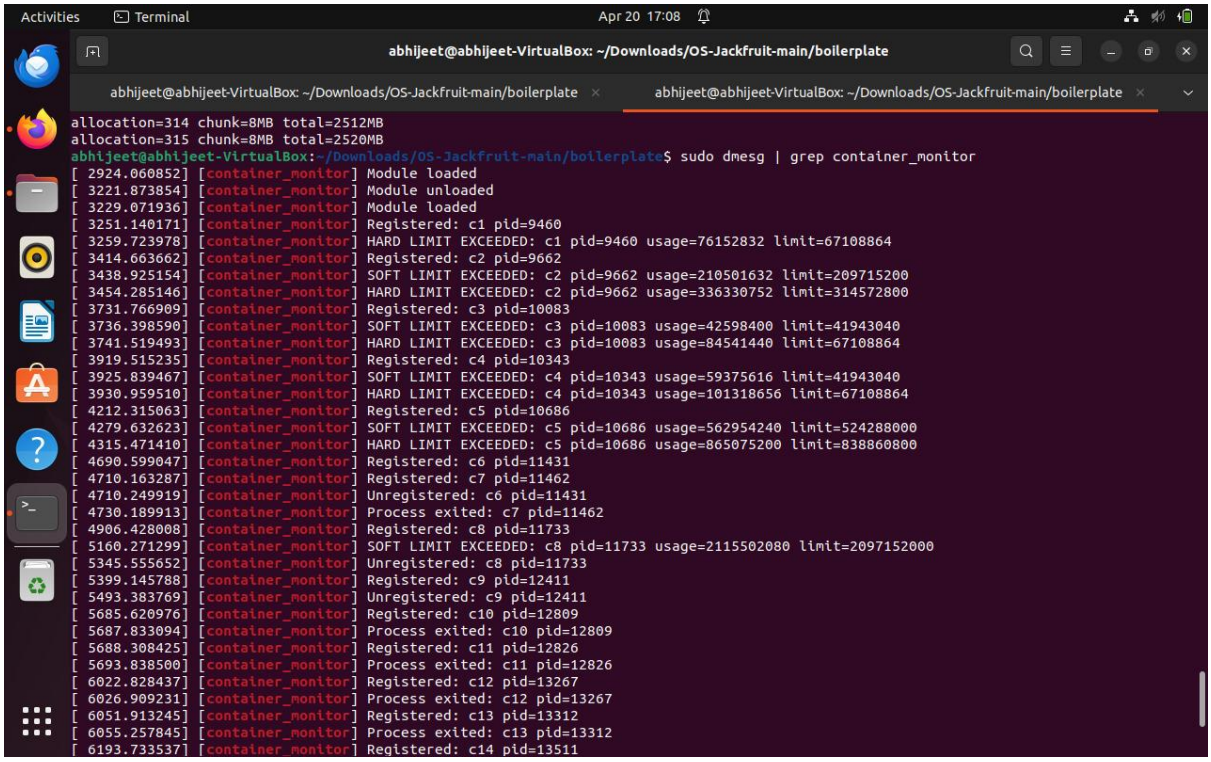
```
Activities   Terminal   Apr 20 17:07
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate   abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine logs s1
allocation=1 chunk=8MB total=8MB
allocation=2 chunk=8MB total=16MB
allocation=3 chunk=8MB total=24MB
allocation=4 chunk=8MB total=32MB
allocation=5 chunk=8MB total=40MB
allocation=6 chunk=8MB total=48MB
allocation=7 chunk=8MB total=56MB
allocation=8 chunk=8MB total=64MB
allocation=9 chunk=8MB total=72MB
allocation=10 chunk=8MB total=80MB
allocation=11 chunk=8MB total=88MB
allocation=12 chunk=8MB total=96MB
allocation=13 chunk=8MB total=104MB
allocation=14 chunk=8MB total=112MB
allocation=15 chunk=8MB total=120MB
allocation=16 chunk=8MB total=128MB
allocation=17 chunk=8MB total=136MB
allocation=18 chunk=8MB total=144MB
allocation=19 chunk=8MB total=152MB
allocation=20 chunk=8MB total=160MB
allocation=21 chunk=8MB total=168MB
allocation=22 chunk=8MB total=176MB
allocation=23 chunk=8MB total=184MB
allocation=24 chunk=8MB total=192MB
allocation=25 chunk=8MB total=200MB
allocation=26 chunk=8MB total=208MB
allocation=27 chunk=8MB total=216MB
allocation=28 chunk=8MB total=224MB
allocation=29 chunk=8MB total=232MB
allocation=30 chunk=8MB total=240MB
allocation=31 chunk=8MB total=248MB
allocation=32 chunk=8MB total=256MB
allocation=33 chunk=8MB total=264MB
allocation=34 chunk=8MB total=272MB
allocation=35 chunk=8MB total=280MB
allocation=36 chunk=8MB total=288MB
allocation=37 chunk=8MB total=296MB
```

3.1


Activities
Terminal
Apr 20 17:07
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
allocation=272 chunk=8MB total=2176MB
allocation=273 chunk=8MB total=2184MB
allocation=274 chunk=8MB total=2192MB
allocation=275 chunk=8MB total=2200MB
allocation=276 chunk=8MB total=2208MB
allocation=277 chunk=8MB total=2216MB
allocation=278 chunk=8MB total=2224MB
allocation=279 chunk=8MB total=2232MB
allocation=280 chunk=8MB total=2240MB
allocation=281 chunk=8MB total=2248MB
allocation=282 chunk=8MB total=2256MB
allocation=283 chunk=8MB total=2264MB
allocation=284 chunk=8MB total=2272MB
allocation=285 chunk=8MB total=2280MB
allocation=286 chunk=8MB total=2288MB
allocation=287 chunk=8MB total=2296MB
allocation=288 chunk=8MB total=2304MB
allocation=289 chunk=8MB total=2312MB
allocation=290 chunk=8MB total=2320MB
allocation=291 chunk=8MB total=2328MB
allocation=292 chunk=8MB total=2336MB
allocation=293 chunk=8MB total=2344MB
allocation=294 chunk=8MB total=2352MB
allocation=295 chunk=8MB total=2360MB
allocation=296 chunk=8MB total=2368MB
allocation=297 chunk=8MB total=2376MB
allocation=298 chunk=8MB total=2384MB
allocation=299 chunk=8MB total=2392MB
allocation=300 chunk=8MB total=2400MB
allocation=301 chunk=8MB total=2408MB
allocation=302 chunk=8MB total=2416MB
allocation=303 chunk=8MB total=2424MB
allocation=304 chunk=8MB total=2432MB
allocation=305 chunk=8MB total=2440MB
allocation=306 chunk=8MB total=2448MB
allocation=307 chunk=8MB total=2456MB
allocation=308 chunk=8MB total=2464MB
allocation=309 chunk=8MB total=2472MB

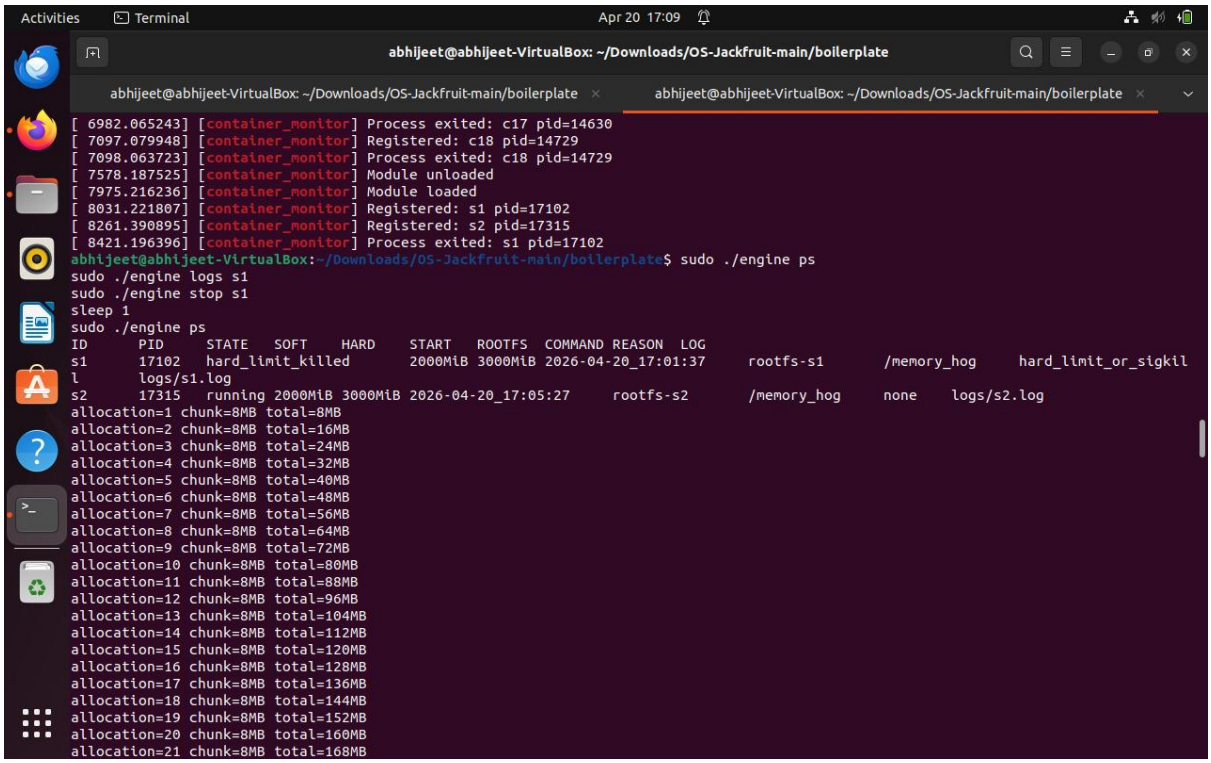
3.2


Activities
Terminal
Apr 20 17:07
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
allocation=272 chunk=8MB total=2176MB
allocation=273 chunk=8MB total=2184MB
allocation=274 chunk=8MB total=2192MB
allocation=275 chunk=8MB total=2200MB
allocation=276 chunk=8MB total=2208MB
allocation=277 chunk=8MB total=2216MB
allocation=278 chunk=8MB total=2224MB
allocation=279 chunk=8MB total=2232MB
allocation=280 chunk=8MB total=2240MB
allocation=281 chunk=8MB total=2248MB
allocation=282 chunk=8MB total=2256MB
allocation=283 chunk=8MB total=2264MB
allocation=284 chunk=8MB total=2272MB
allocation=285 chunk=8MB total=2280MB
allocation=286 chunk=8MB total=2288MB
allocation=287 chunk=8MB total=2296MB
allocation=288 chunk=8MB total=2304MB
allocation=289 chunk=8MB total=2312MB
allocation=290 chunk=8MB total=2320MB
allocation=291 chunk=8MB total=2328MB
allocation=292 chunk=8MB total=2336MB
allocation=293 chunk=8MB total=2344MB
allocation=294 chunk=8MB total=2352MB
allocation=295 chunk=8MB total=2360MB
allocation=296 chunk=8MB total=2368MB
allocation=297 chunk=8MB total=2376MB
allocation=298 chunk=8MB total=2384MB
allocation=299 chunk=8MB total=2392MB
allocation=300 chunk=8MB total=2400MB
allocation=301 chunk=8MB total=2408MB
allocation=302 chunk=8MB total=2416MB
allocation=303 chunk=8MB total=2424MB
allocation=304 chunk=8MB total=2432MB
allocation=305 chunk=8MB total=2440MB
allocation=306 chunk=8MB total=2448MB
allocation=307 chunk=8MB total=2456MB
allocation=308 chunk=8MB total=2464MB
allocation=309 chunk=8MB total=2472MB

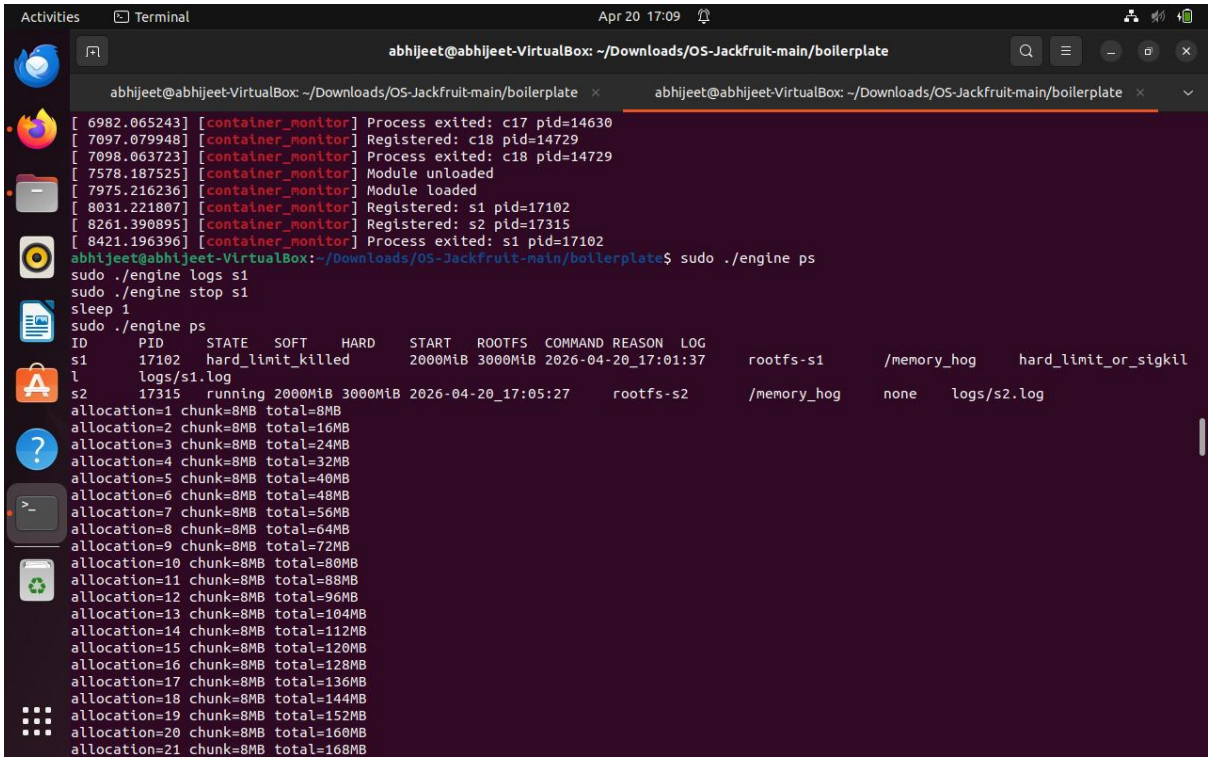


```
allocation=314 chunk=8MB total=2512MB
allocation=315 chunk=8MB total=2520MB
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo dmesg | grep container_monitor
[ 2924.060852] [container_monitor] Module loaded
[ 3221.873854] [container_monitor] Module unloaded
[ 3229.071936] [container_monitor] Module loaded
[ 3251.140171] [container_monitor] Registered: c1 pid=9460
[ 3259.723978] [container_monitor] HARD LIMIT EXCEEDED: c1 pid=9460 usage=76152832 limit=67108864
[ 3414.663662] [container_monitor] Registered: c2 pid=9662
[ 3438.925154] [container_monitor] SOFT LIMIT EXCEEDED: c2 pid=9662 usage=210501632 limit=209715200
[ 3454.285146] [container_monitor] HARD LIMIT EXCEEDED: c2 pid=9662 usage=336330752 limit=314572800
[ 3731.766909] [container_monitor] Registered: c3 pid=10083
[ 3736.398590] [container_monitor] SOFT LIMIT EXCEEDED: c3 pid=10083 usage=42598400 limit=41943040
[ 3741.519493] [container_monitor] HARD LIMIT EXCEEDED: c3 pid=10083 usage=84541440 limit=67108864
[ 3919.515235] [container_monitor] Registered: c4 pid=10343
[ 3925.839467] [container_monitor] SOFT LIMIT EXCEEDED: c4 pid=10343 usage=59375616 limit=41943040
[ 3930.959510] [container_monitor] HARD LIMIT EXCEEDED: c4 pid=10343 usage=101318656 limit=67108864
[ 4212.315063] [container_monitor] Registered: c5 pid=10686
[ 4279.632623] [container_monitor] SOFT LIMIT EXCEEDED: c5 pid=10686 usage=562954240 limit=524288000
[ 4315.471410] [container_monitor] HARD LIMIT EXCEEDED: c5 pid=10686 usage=865075200 limit=838860800
[ 4690.599047] [container_monitor] Registered: c6 pid=11431
[ 4710.163287] [container_monitor] Registered: c7 pid=11462
[ 4710.249919] [container_monitor] Unregistered: c6 pid=11431
[ 4730.189913] [container_monitor] Process exited: c7 pid=11462
[ 4906.428008] [container_monitor] Registered: c8 pid=11733
[ 5160.271299] [container_monitor] SOFT LIMIT EXCEEDED: c8 pid=11733 usage=2115502080 limit=2097152000
[ 5345.555652] [container_monitor] Unregistered: c8 pid=11733
[ 5399.145788] [container_monitor] Registered: c9 pid=12411
[ 5493.383769] [container_monitor] Unregistered: c9 pid=12411
[ 5685.620976] [container_monitor] Registered: c10 pid=12809
[ 5687.833094] [container_monitor] Process exited: c10 pid=12809
[ 5688.308425] [container_monitor] Registered: c11 pid=12826
[ 5693.838500] [container_monitor] Process exited: c11 pid=12826
[ 6022.828437] [container_monitor] Registered: c12 pid=13267
[ 6026.909231] [container_monitor] Process exited: c12 pid=13267
[ 6051.913245] [container_monitor] Registered: c13 pid=13312
[ 6055.257845] [container_monitor] Process exited: c13 pid=13312
[ 6193.733537] [container_monitor] Registered: c14 pid=13511
```

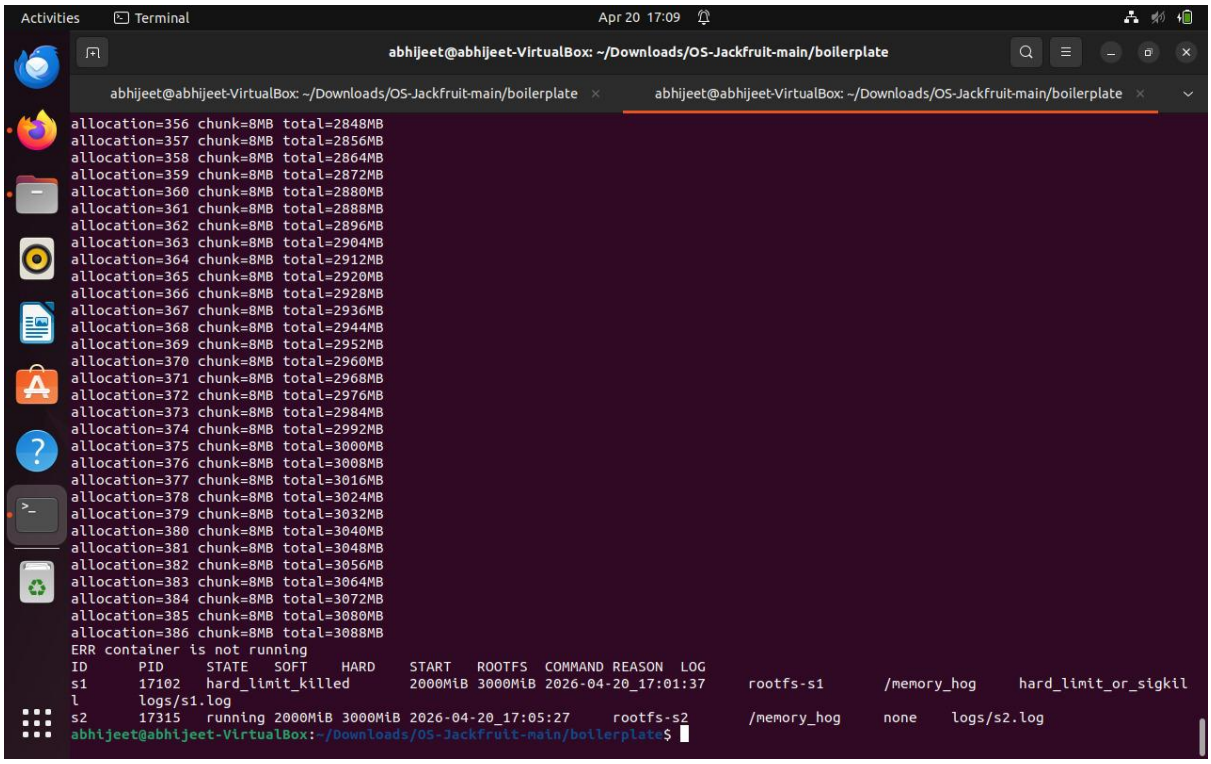
4CLI-IPC



```
[ 6982.065243] [container_monitor] Process exited: c17 pid=14630
[ 7097.079948] [container_monitor] Registered: c18 pid=14729
[ 7098.063723] [container_monitor] Process exited: c18 pid=14729
[ 7578.187525] [container_monitor] Module unloaded
[ 7975.216236] [container_monitor] Module loaded
[ 8031.221807] [container_monitor] Registered: s1 pid=17102
[ 8261.390895] [container_monitor] Registered: s2 pid=17315
[ 8421.196396] [container_monitor] Process exited: s1 pid=17102
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine ps
sudo ./engine logs s1
sudo ./engine stop s1
sleep 1
sudo ./engine ps
ID      PID     STATE   SOFT    HARD    START   ROOTFS  COMMAND REASON  LOG
s1      17102   hard_limit_killed       2000MiB 3000MiB 2026-04-20_17:01:37     rootfs-s1       /memory_hog     hard_limit_or_sigkil
l       logs/s1.log
s2      17315   running 2000MiB 3000MiB 2026-04-20_17:05:27     rootfs-s2       /memory_hog     none    logs/s2.log
allocation=1 chunk=8MB total=8MB
allocation=2 chunk=8MB total=16MB
allocation=3 chunk=8MB total=24MB
allocation=4 chunk=8MB total=32MB
allocation=5 chunk=8MB total=40MB
allocation=6 chunk=8MB total=48MB
allocation=7 chunk=8MB total=56MB
allocation=8 chunk=8MB total=64MB
allocation=9 chunk=8MB total=72MB
allocation=10 chunk=8MB total=80MB
allocation=11 chunk=8MB total=88MB
allocation=12 chunk=8MB total=96MB
allocation=13 chunk=8MB total=104MB
allocation=14 chunk=8MB total=112MB
allocation=15 chunk=8MB total=120MB
allocation=16 chunk=8MB total=128MB
allocation=17 chunk=8MB total=136MB
allocation=18 chunk=8MB total=144MB
allocation=19 chunk=8MB total=152MB
allocation=20 chunk=8MB total=160MB
allocation=21 chunk=8MB total=168MB
```



5.SOFT-LIMIT WARNING



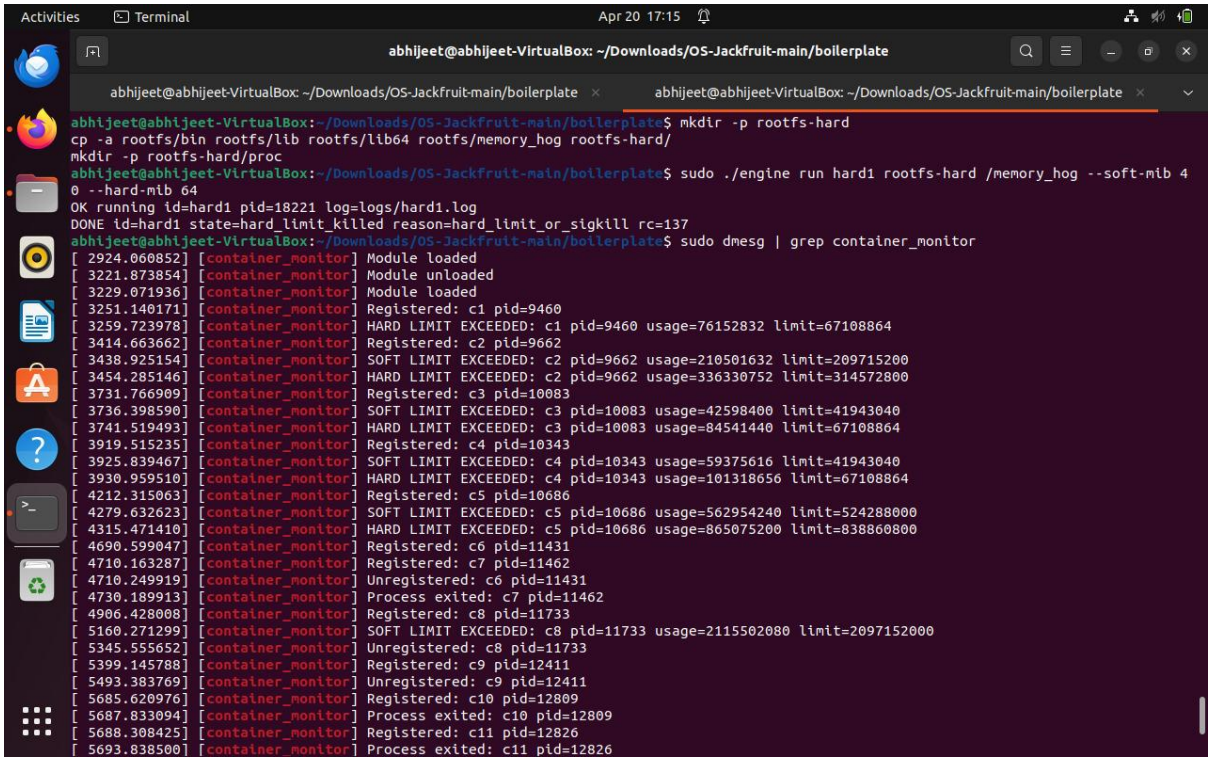
```
Activities  Terminal  Apr 20 17:11
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate   abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
[ 4690.599047] [container_monitor] Registered: c6 pid=11431
[ 4710.163287] [container_monitor] Registered: c7 pid=11462
[ 4710.249919] [container_monitor] Unregistered: c6 pid=11431
[ 4730.189913] [container_monitor] Process exited: c7 pid=11462
[ 4906.428008] [container_monitor] Registered: c8 pid=11733
[ 5160.271299] [container_monitor] SOFT LIMIT EXCEEDED: c8 pid=11733 usage=2115502080 limit=2097152000
[ 5345.555652] [container_monitor] Unregistered: c8 pid=11733
[ 5399.145788] [container_monitor] Registered: c9 pid=12411
[ 5493.383769] [container_monitor] Unregistered: c9 pid=12411
[ 5685.620976] [container_monitor] Registered: c10 pid=12809
[ 5687.833094] [container_monitor] Process exited: c10 pid=12809
[ 5688.308425] [container_monitor] Registered: c11 pid=12826
[ 5693.838500] [container_monitor] Process exited: c11 pid=12826
[ 6022.828437] [container_monitor] Registered: c12 pid=13267
[ 6026.909231] [container_monitor] Process exited: c12 pid=13267
[ 6051.913245] [container_monitor] Registered: c13 pid=13312
[ 6055.257845] [container_monitor] Process exited: c13 pid=13312
[ 6193.733537] [container_monitor] Registered: c14 pid=13511
[ 6197.937766] [container_monitor] Process exited: c14 pid=13511
[ 6518.389911] [container_monitor] Registered: c15 pid=14051
[ 6638.157394] [container_monitor] Registered: c16 pid=14196
[ 6638.172054] [container_monitor] Unregistered: c15 pid=14051
[ 6722.468057] [container_monitor] Process exited: c16 pid=14196
[ 6976.765386] [container_monitor] Registered: c17 pid=14630
[ 6982.065243] [container_monitor] Process exited: c17 pid=14630
[ 7097.079948] [container_monitor] Registered: c18 pid=14729
[ 7098.063723] [container_monitor] Process exited: c18 pid=14729
[ 7578.187525] [container_monitor] Module unloaded
[ 7975.216236] [container_monitor] Module loaded
[ 8031.221807] [container_monitor] Registered: s1 pid=17102
[ 8261.390895] [container_monitor] Registered: s2 pid=17315
[ 8421.196396] [container_monitor] Process exited: s1 pid=17102
[ 8544.530600] [container_monitor] Registered: soft1 pid=17826
[ 8549.199227] [container_monitor] SOFT LIMIT EXCEEDED: soft1 pid=17826 usage=42598400 limit=41943040
[ 8549.199243] [container_monitor] SOFT LIMIT EXCEEDED: s2 pid=17315 usage=2114977792 limit=2097152000
[ 8569.679784] [container_monitor] HARD LIMIT EXCEEDED: soft1 pid=17826 usage=217579520 limit=209715200
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$
```

6.HARD LIMIT ENFORCEMENT

```
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ mkdir -p rootfs-hard
cp -a rootfs/bin rootfs/lib rootfs/lib64 rootfs/memory_hog rootfs-hard/
mkdir -p rootfs-hard/proc
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine run hard1 rootfs-hard /memory_hog --soft-mib 4
0 --hard-mib 64
OK running id=hard1 pid=18221 log=logs/hard1.log
DONE id=hard1 state=hard_limit_killed reason=hard_limit_or_sigkill rc=137
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$
```

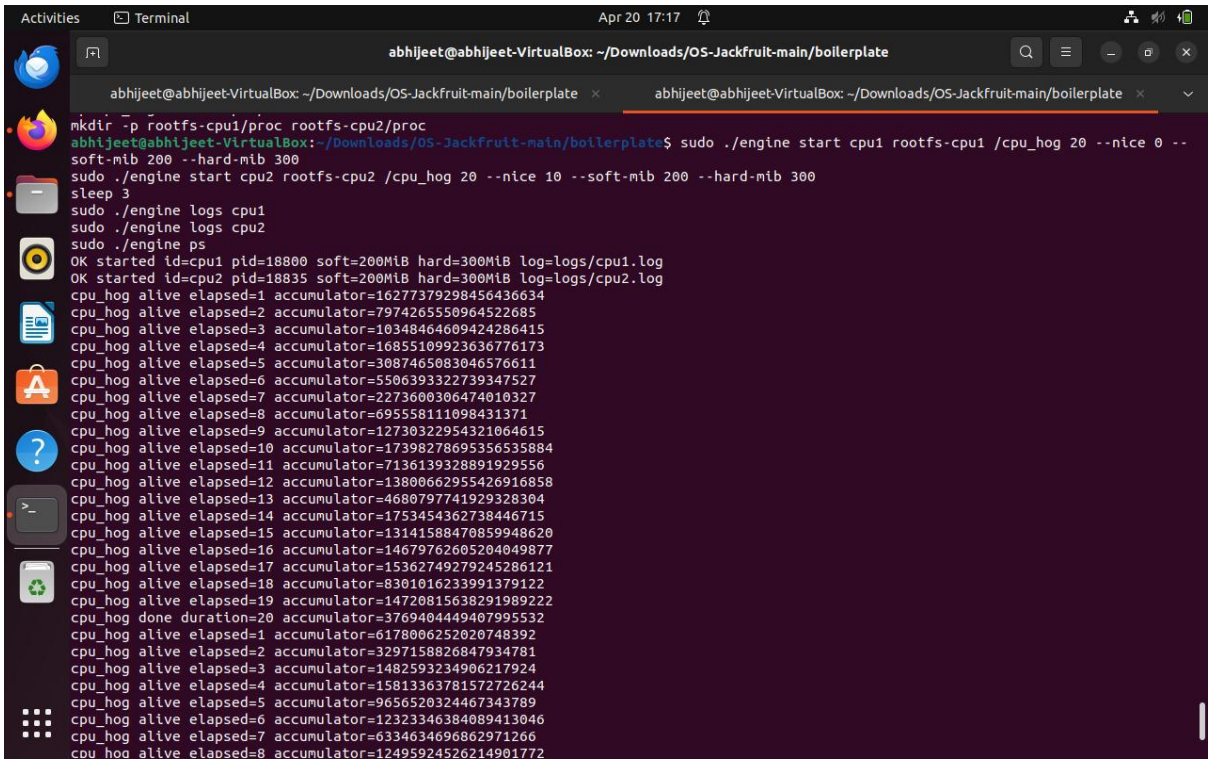
6.1

```
Activities  Terminal  Apr 20 17:15
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate   abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ mkdir -p rootfs-hard
cp -a rootfs/bin rootfs/lib rootfs/lib64 rootfs/memory_hog rootfs-hard/
mkdir -p rootfs-hard/proc
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine run hard1 rootfs-hard /memory_hog --soft-mib 4
0 --hard-mib 64
OK running id=hard1 pid=18221 log=logs/hard1.log
DONE id=hard1 state=hard_limit_killed reason=hard_limit_or_sigkill rc=137
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo dmesg | grep container_monitor
[ 2924.060852] [container_monitor] Module loaded
[ 3221.873854] [container_monitor] Module unloaded
[ 3229.071936] [container_monitor] Module loaded
[ 3251.140171] [container_monitor] Registered: c1 pid=9460
[ 3259.723978] [container_monitor] HARD LIMIT EXCEEDED: c1 pid=9460 usage=76152832 limit=67108864
[ 3414.663662] [container_monitor] Registered: c2 pid=9662
[ 3438.925154] [container_monitor] SOFT LIMIT EXCEEDED: c2 pid=9662 usage=210501632 limit=209715200
[ 3454.285146] [container_monitor] HARD LIMIT EXCEEDED: c2 pid=9662 usage=336330752 limit=314572800
[ 3731.766909] [container_monitor] Registered: c3 pid=10083
[ 3736.398590] [container_monitor] SOFT LIMIT EXCEEDED: c3 pid=10083 usage=42598400 limit=41943040
[ 3741.519493] [container_monitor] HARD LIMIT EXCEEDED: c3 pid=10083 usage=84541440 limit=67108864
[ 3919.515235] [container_monitor] Registered: c4 pid=10343
[ 3925.839467] [container_monitor] SOFT LIMIT EXCEEDED: c4 pid=10343 usage=59375616 limit=41943040
[ 3930.959510] [container_monitor] HARD LIMIT EXCEEDED: c4 pid=10343 usage=101318656 limit=67108864
[ 4212.315063] [container_monitor] Registered: c5 pid=10686
[ 4279.632623] [container_monitor] SOFT LIMIT EXCEEDED: c5 pid=10686 usage=562954240 limit=524288000
[ 4315.471410] [container_monitor] HARD LIMIT EXCEEDED: c5 pid=10686 usage=865075200 limit=838860800
[ 4690.599047] [container_monitor] Registered: c6 pid=11431
[ 4710.163287] [container_monitor] Registered: c7 pid=11462
[ 4710.249919] [container_monitor] Unregistered: c6 pid=11431
[ 4730.189913] [container_monitor] Process exited: c7 pid=11462
[ 4906.428008] [container_monitor] Registered: c8 pid=11733
[ 5160.271299] [container_monitor] SOFT LIMIT EXCEEDED: c8 pid=11733 usage=2115502080 limit=2097152000
[ 5345.555652] [container_monitor] Unregistered: c8 pid=11733
[ 5399.145788] [container_monitor] Registered: c9 pid=12411
[ 5493.383769] [container_monitor] Unregistered: c9 pid=12411
[ 5685.620976] [container_monitor] Registered: c10 pid=12809
[ 5687.833094] [container_monitor] Process exited: c10 pid=12809
[ 5688.308425] [container_monitor] Registered: c11 pid=12826
[ 5693.838500] [container_monitor] Process exited: c11 pid=12826
```



```
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ mkdir -p rootfs-hard
cp -a rootfs/bin rootfs/lib rootfs/lib64 rootfs/memory_hog rootfs-hard/
mkdir -p rootfs-hard/proc
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine run hard1 rootfs-hard /memory_hog --soft-mib 4
0 --hard-mib 64
OK running id=hard1 pid=18221 log=logs/hard1.log
DONE id=hard1 state=hard_limit_killed reason=hard_limit_or_sigkill rc=137
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo dmesg | grep container_monitor
[ 2924.060852] [container_monitor] Module loaded
[ 3221.873854] [container_monitor] Module unloaded
[ 3229.071936] [container_monitor] Module loaded
[ 3251.140171] [container_monitor] Registered: c1 pid=9460
[ 3259.723978] [container_monitor] HARD LIMIT EXCEEDED: c1 pid=9460 usage=76152832 limit=67108864
[ 3414.663662] [container_monitor] Registered: c2 pid=9662
[ 3438.925154] [container_monitor] SOFT LIMIT EXCEEDED: c2 pid=9662 usage=210501632 limit=209715200
[ 3454.285146] [container_monitor] HARD LIMIT EXCEEDED: c2 pid=9662 usage=336330752 limit=314572800
[ 3731.766909] [container_monitor] Registered: c3 pid=10083
[ 3736.398590] [container_monitor] SOFT LIMIT EXCEEDED: c3 pid=10083 usage=42598400 limit=41943040
[ 3741.519493] [container_monitor] HARD LIMIT EXCEEDED: c3 pid=10083 usage=84541440 limit=67108864
[ 3919.515235] [container_monitor] Registered: c4 pid=10343
[ 3925.839467] [container_monitor] SOFT LIMIT EXCEEDED: c4 pid=10343 usage=59375616 limit=41943040
[ 3930.959510] [container_monitor] HARD LIMIT EXCEEDED: c4 pid=10343 usage=101318656 limit=67108864
[ 4212.315063] [container_monitor] Registered: c5 pid=10686
[ 4279.632623] [container_monitor] SOFT LIMIT EXCEEDED: c5 pid=10686 usage=562954240 limit=524288000
[ 4315.471410] [container_monitor] HARD LIMIT EXCEEDED: c5 pid=10686 usage=865075200 limit=838860800
[ 4690.599047] [container_monitor] Registered: c6 pid=11431
[ 4710.163287] [container_monitor] Registered: c7 pid=11462
[ 4710.249919] [container_monitor] Unregistered: c6 pid=11431
[ 4730.189913] [container_monitor] Process exited: c7 pid=11462
[ 4906.428008] [container_monitor] Registered: c8 pid=11733
[ 5160.271299] [container_monitor] SOFT LIMIT EXCEEDED: c8 pid=11733 usage=2115502080 limit=2097152000
[ 5345.555652] [container_monitor] Unregistered: c8 pid=11733
[ 5399.145788] [container_monitor] Registered: c9 pid=12411
[ 5493.383769] [container_monitor] Unregistered: c9 pid=12411
[ 5685.620976] [container_monitor] Registered: c10 pid=12809
[ 5687.833094] [container_monitor] Process exited: c10 pid=12809
[ 5688.308425] [container_monitor] Registered: c11 pid=12826
[ 5693.838500] [container_monitor] Process exited: c11 pid=12826
```

7.SCHEDULING EXPERIMENT



```
mkdir -p rootfs-cpu1/proc rootfs-cpu2/proc
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine start cpu1 rootfs-cpu1 /cpu_hog 20 --nice 0 --
soft-mib 200 --hard-mib 300
sudo ./engine start cpu2 rootfs-cpu2 /cpu_hog 20 --nice 10 --soft-mib 200 --hard-mib 300
sleep 3
sudo ./engine logs cpu1
sudo ./engine logs cpu2
sudo ./engine ps
OK started id=cpu1 pid=18800 soft=200MiB hard=300MiB log=logs/cpu1.log
OK started id=cpu2 pid=18835 soft=200MiB hard=300MiB log=logs/cpu2.log
cpu_hog alive elapsed=1 accumulator=16277379298456436634
cpu_hog alive elapsed=2 accumulator=7974265550964522685
cpu_hog alive elapsed=3 accumulator=10348464609424286415
cpu_hog alive elapsed=4 accumulator=16855109923636776173
cpu_hog alive elapsed=5 accumulator=3087465083046576611
cpu_hog alive elapsed=6 accumulator=5506393322739347527
cpu_hog alive elapsed=7 accumulator=2273600306474010327
cpu_hog alive elapsed=8 accumulator=695558111098431371
cpu_hog alive elapsed=9 accumulator=12730322954321064615
cpu_hog alive elapsed=10 accumulator=17398278695356535884
cpu_hog alive elapsed=11 accumulator=7136139328891929556
cpu_hog alive elapsed=12 accumulator=13800662955426916858
cpu_hog alive elapsed=13 accumulator=4680797741929328304
cpu_hog alive elapsed=14 accumulator=1753454362738446715
cpu_hog alive elapsed=15 accumulator=13141588470859948620
cpu_hog alive elapsed=16 accumulator=14679762605204049877
cpu_hog alive elapsed=17 accumulator=15362749279245286121
cpu_hog alive elapsed=18 accumulator=8301016233991379122
cpu_hog alive elapsed=19 accumulator=14720815638291989222
cpu_hog done duration=20 accumulator=3769404449407995532
cpu_hog alive elapsed=1 accumulator=6178006252020748392
cpu_hog alive elapsed=2 accumulator=3297158826847934781
cpu_hog alive elapsed=3 accumulator=1482593234906217924
cpu_hog alive elapsed=4 accumulator=15813363781572726244
cpu_hog alive elapsed=5 accumulator=9656520324467343789
cpu_hog alive elapsed=6 accumulator=12323346384089413046
cpu_hog alive elapsed=7 accumulator=6334634696862971266
cpu_hog alive elapsed=8 accumulator=12495924526214901772
```

```
Activities   Terminal                    Apr 20 17:17
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate   abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
cpu_hog alive elapsed=16 accumulator=14679762605204049877
cpu_hog alive elapsed=17 accumulator=15362749279245286121
cpu_hog alive elapsed=18 accumulator=8301016233991379122
cpu_hog alive elapsed=19 accumulator=14720815638291989222
cpu_hog done duration=20 accumulator=3769404449407995532
cpu_hog alive elapsed=1 accumulator=6178006252020748392
cpu_hog alive elapsed=2 accumulator=3297158826847934781
cpu_hog alive elapsed=3 accumulator=1482593234906217924
cpu_hog alive elapsed=4 accumulator=15813363781572726244
cpu_hog alive elapsed=5 accumulator=9656520324467343789
cpu_hog alive elapsed=6 accumulator=12323346384089413046
cpu_hog alive elapsed=7 accumulator=6334634696862971266
cpu_hog alive elapsed=8 accumulator=12495924526214901772
cpu_hog alive elapsed=9 accumulator=5712632720580074087
cpu_hog alive elapsed=10 accumulator=2314503336099310226
cpu_hog alive elapsed=11 accumulator=3882063253450999942
cpu_hog alive elapsed=12 accumulator=9821250246082978185
cpu_hog alive elapsed=13 accumulator=7815347916348721671
cpu_hog alive elapsed=14 accumulator=9849353884185488701
cpu_hog alive elapsed=15 accumulator=1789365546135315295
cpu_hog alive elapsed=16 accumulator=14721869783360271631
cpu_hog alive elapsed=17 accumulator=7881904373987522341
cpu_hog alive elapsed=18 accumulator=11022118738145272239
cpu_hog alive elapsed=19 accumulator=8965151310516402924
cpu_hog done duration=20 accumulator=5090431493933811102
ID      PID     STATE   SOFT    HARD    START   ROOTFS  COMMAND REASON  LOG
s1      17102   hard_limit_killed       2000MiB 3000MiB 2026-04-20_17:01:37     rootfs-s1       /memory_hog     hard_limit_or_sigkil
l       logs/s1.log
s2      17315   hard_limit_killed       2000MiB 3000MiB 2026-04-20_17:05:27     rootfs-s2       /memory_hog     hard_limit_or_sigkil
l       logs/s2.log
soft1   17826   hard_limit_killed       40MiB   200MiB  2026-04-20_17:10:10     rootfs-soft     /memory_hog     hard_limit_or_sigkil
l       logs/soft1.log
hard1   18221   hard_limit_killed       40MiB   64MiB   2026-04-20_17:12:38     rootfs-hard     /memory_hog     hard_limit_or_sigkil
l       logs/hard1.log
cpu1    18800   exited  200MiB  300MiB  2026-04-20_17:16:09     rootfs-cpu1     /cpu_hog 20     exit_code_0     logs/cpu1.log
cpu2    18835   exited  200MiB  300MiB  2026-04-20_17:16:29     rootfs-cpu2     /cpu_hog 20     exit_code_0     logs/cpu2.log
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$
```

8.CLEAR TEARDOWN

```
Activities   Terminal                    Apr 20 17:17
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate   abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
cpu_hog alive elapsed=16 accumulator=14679762605204049877
cpu_hog alive elapsed=17 accumulator=15362749279245286121
cpu_hog alive elapsed=18 accumulator=8301016233991379122
cpu_hog alive elapsed=19 accumulator=14720815638291989222
cpu_hog done duration=20 accumulator=3769404449407995532
cpu_hog alive elapsed=1 accumulator=6178006252020748392
cpu_hog alive elapsed=2 accumulator=3297158826847934781
cpu_hog alive elapsed=3 accumulator=1482593234906217924
cpu_hog alive elapsed=4 accumulator=15813363781572726244
cpu_hog alive elapsed=5 accumulator=9656520324467343789
cpu_hog alive elapsed=6 accumulator=12323346384089413046
cpu_hog alive elapsed=7 accumulator=6334634696862971266
cpu_hog alive elapsed=8 accumulator=12495924526214901772
cpu_hog alive elapsed=9 accumulator=5712632720580074087
cpu_hog alive elapsed=10 accumulator=2314503336099310226
cpu_hog alive elapsed=11 accumulator=3882063253450999942
cpu_hog alive elapsed=12 accumulator=9821250246082978185
cpu_hog alive elapsed=13 accumulator=7815347916348721671
cpu_hog alive elapsed=14 accumulator=9849353884185488701
cpu_hog alive elapsed=15 accumulator=1789365546135315295
cpu_hog alive elapsed=16 accumulator=14721869783360271631
cpu_hog alive elapsed=17 accumulator=7881904373987522341
cpu_hog alive elapsed=18 accumulator=11022118738145272239
cpu_hog alive elapsed=19 accumulator=8965151310516402924
cpu_hog done duration=20 accumulator=5090431493933811102
ID      PID     STATE   SOFT    HARD    START   ROOTFS  COMMAND REASON  LOG
s1      17102   hard_limit_killed       2000MiB 3000MiB 2026-04-20_17:01:37     rootfs-s1       /memory_hog     hard_limit_or_sigkil
l       logs/s1.log
s2      17315   hard_limit_killed       2000MiB 3000MiB 2026-04-20_17:05:27     rootfs-s2       /memory_hog     hard_limit_or_sigkil
l       logs/s2.log
soft1   17826   hard_limit_killed       40MiB   200MiB  2026-04-20_17:10:10     rootfs-soft     /memory_hog     hard_limit_or_sigkil
l       logs/soft1.log
hard1   18221   hard_limit_killed       40MiB   64MiB   2026-04-20_17:12:38     rootfs-hard     /memory_hog     hard_limit_or_sigkil
l       logs/hard1.log
cpu1    18800   exited  200MiB  300MiB  2026-04-20_17:16:09     rootfs-cpu1     /cpu_hog 20     exit_code_0     logs/cpu1.log
cpu2    18835   exited  200MiB  300MiB  2026-04-20_17:16:29     rootfs-cpu2     /cpu_hog 20     exit_code_0     logs/cpu2.log
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$
```

```
Activities   Terminal                    Apr 20 17:20
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate   abhijeet@abhijeet-VirtualBox: ~/Downloads/OS-Jackfruit-main/boilerplate
cpu2    18835   exited  200MiB  300MiB  2026-04-20_17:16:29     rootfs-cpu2     /cpu_hog 20     exit_code_0     logs/cpu2.log
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine stop s2
sudo ./engine stop soft1
sudo ./engine stop cpu1
sudo ./engine stop cpu2
sleep 1
ERR container is not running
ERR container is not running
ERR container is not running
ERR container is not running
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo ./engine shutdown
OK supervisor shutting down
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ ps aux | grep defunct
abhijeet   19111  0.0  0.0   9080  2612 pts/2    S+   17:18   0:00 grep --color=auto defunct
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo rmmod monitor
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ v
v: command not found
abhijeet@abhijeet-VirtualBox:~/Downloads/OS-Jackfruit-main/boilerplate$ sudo dmesg | grep container_monitor
[ 2924.060852] [container_monitor] Module loaded
[ 3221.873854] [container_monitor] Module unloaded
[ 3229.071936] [container_monitor] Module loaded
[ 3251.140171] [container_monitor] Registered: c1 pid=9460
[ 3259.723978] [container_monitor] HARD LIMIT EXCEEDED: c1 pid=9460 usage=76152832 limit=67108864
[ 3414.663662] [container_monitor] Registered: c2 pid=9662
[ 3438.925154] [container_monitor] SOFT LIMIT EXCEEDED: c2 pid=9662 usage=210501632 limit=209715200
[ 3454.285146] [container_monitor] HARD LIMIT EXCEEDED: c2 pid=9662 usage=336330752 limit=314572800
[ 3731.766909] [container_monitor] Registered: c3 pid=10083
[ 3736.398590] [container_monitor] SOFT LIMIT EXCEEDED: c3 pid=10083 usage=42598400 limit=41943040
[ 3741.519493] [container_monitor] HARD LIMIT EXCEEDED: c3 pid=10083 usage=84541440 limit=67108864
[ 3919.515235] [container_monitor] Registered: c4 pid=10343
[ 3925.839467] [container_monitor] SOFT LIMIT EXCEEDED: c4 pid=10343 usage=59375616 limit=41943040
[ 3930.959510] [container_monitor] HARD LIMIT EXCEEDED: c4 pid=10343 usage=101318656 limit=67108864
[ 4212.315063] [container_monitor] Registered: c5 pid=10686
[ 4279.632623] [container_monitor] SOFT LIMIT EXCEEDED: c5 pid=10686 usage=562954240 limit=524288000
[ 4315.471410] [container_monitor] HARD LIMIT EXCEEDED: c5 pid=10686 usage=865075200 limit=838860800
[ 4690.599047] [container_monitor] Registered: c6 pid=11431
[ 4710.163287] [container_monitor] Registered: c7 pid=11462
[ 4710.249919] [container_monitor] Unregistered: c6 pid=11431
```